Multiscanning Strategy-Based Recurrent Neural Network for Hyperspectral Image Classification

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***Abstract*— Most methods based on the convolutional neural network show satisfying performance for hyperspectral image (HSI) classification. However, the spatial dependence among different pixels is not well learned by CNNs. A recurrent neural network (RNN) can effectively establish the dependence of nonadjacent pixels and ensure that each feature activation in its output is an activation at the specific location concerning the whole image, in contrast to the usual local context window in the CNNs. However, recent limited conversion schemes in RNN-based methods for HSI classification cannot fully capture the complete spatial dependence of an HSI patch. In this study, a novel multiscanning strategy with RNN is proposed to feature the sequential character of the HSI pixel and fully consider the spatial dependence in the HSI patch. By investigating different scanning forms, eight scanning orders are considered spatially, which flattens one local HSI patch into eight neighboring continuous pixel sequences. Moreover, considering that eight scanning orders complement one local patch with correlative dependence, the concatenated features from all scanning orders are fed into the RNN again for complementarity. As a result, the network can achieve competitive classification performance on three publicly accessible datasets using fewer parameters than other state-of-the-art methods.**

***Index Terms*— Deep learning, hyperspectral image classification, multiscanning strategy, recurrent neural network.**

I. Introduction

HYPERSPECTRAL imaging technology is critical in the remote sensing (RS) field due to its high resolution in the spatial and spectral domains [1]. Furthermore, the hyperspectral sensors capture the information from visible to infrared light wavelength in the electromagnetic spectrum, which can help record the chemical or physical properties of different materials [2]. Therefore, the generated image, namely, the hyperspectral image (HSI), which is considered a 3-D data cube, provides compelling and informative data for monitoring environmental detection [3], military inspection [4], mining exploration [5], and medical diagnosis [6]. Therefore, the HSI classification task, which is the first step for these applications that aim to assign a label to each pixel, becomes critical because the high accuracy of HSI classification results will improve the quality of these applications. Hence, recently, this task has been widely researched in the RS field [7], [8].

Deep learning has recently become trendy in RS, especially for HSI classification tasks [9], [10]. Convolutional neural networks (CNNs) are among the most popular deep learning models for dealing with image features. Moreover, CNNs have been widely used for HSI classification research. In [16], [17] and [38], the spatial information in one local patch around a corresponding pixel was implemented by using 2-D CNN after performing the principal component analysis (PCA) operation. Li *et al.* [18] proposed a fully CNN with the first principal component (PC) of HSI through convolution and deconvolution. Lee and Heesung [19] proposed a deep contextual 2-D CNN to exploit the local contextual correlations. In [21], a simple 3-D CNN structure is firstly proposed for HSI classification by integrating spectral information and spatial information. He *et al.* [35] proposed a multiscale 3-D CNN to jointly learn both 2-D multiscale spatial feature and 1-D spectral feature. Sun *et al.* [39] proposed one spectral–spatial attention network (SSAN) with 3-D CNN to capture the features from attention areas. Hang *et al.* [40] proposed an attention-aided 3-D CNN model to focus on more discriminative channels and positions. Furthermore, Cao *et al.* [41] integrated both active learning and CNN into a unified framework for improving the classification performance while reducing the labeling cost. Meanwhile, Liu *et al.* [42] considered the existence of unknown labels in the real data and proposed a multitask deep learning method (MDL4OW) to conduct reconstruction and classification simultaneously.

CNN-based methods have obtained good performance in extracting local features. However, the local contextual features depend on the small kernel size of the filter (usually 3 $\times$ 3), and the spatial dependence between nonadjacent pixels in the image patch is not well learned; for example, the upper left and lower right regions of the image patch somehow do not establish the communication [51]. Therefore, the small kernel size limits the CNN receptive field. Although dilated convolution [30], [31] has been proposed for enlarging the receptive field in convolutional operation, it highly relies on dilation rate, which is not appropriate for the small size of the HSI patch. Moreover, it somehow sacrifices the adjacent pixels' information by convolving nonadjacent pixels to some extent, which weakens the continuity of spatial distribution

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concerning feature extraction. Furthermore, the spatial dependence of the HSI patches may lose in the fully connected layer or the pooling layer in CNN-based methods [37]. Thereby, in [36], the author concluded that spatial dependence of all pixels is also essential to interpret the spatial structure information.

Some methods proposed to address the problems of CNN-based methods draw inspiration from the natural language models that are generally flexible and easily generalized for modeling language dependence [46], such as recurrent neural networks (RNNs). Moreover, RNNs have also been used in video processing, which bridges vision and language together. For example, Zhao *et al.* [47] focused on a hierarchical structure of video data and proposed a hierarchical structure-adaptive RNN (HSA-RNN) for video summarization; meanwhile, in [48], the coattention model-based recurrent neural network (CAM-RNN) was proposed for video caption generation where RNN served as a decoder.

Hence, HSIs, similar to video data and a 3-D data cube, can also cooperate with RNN after proper conversion. First, for the spectral aspect, a pixel (spectral signature) in an HSI is analogous to a word vector [43] because both are distributional representations with unique "meaning" [44]. The pixel conveys the land-cover information, while the word vector conveys the meaning of a word. Second, for the spatial aspect, the local region around a target pixel in an HSI is analogous to the context of a target word in a corpus. With these two analogies, few RNN-based methods have been proposed for finding the spatial (contextual) dependence among pixels in the HSI patch. However, the image data are of higher dimensionality, and the conventional RNNs are not suitable for contextual image learning. When applying RNNs to the image, the critical problem is getting a suitable sequence. Hence, on the image level, in [20], the first PC of one HSI image patch was sliced row by row from top to bottom, and then, each row vector was fed into an RNN. Zuo *et al.* [29], [52] scanned the image from four independent directions in row-by-row and column-by-column manners; one row or one column will be one input for the RNN unit, and then, they were summed together for the final representations. On the pixel level, in [33], one HSI image patch was sorted into one sequence by each pixel's importance; one step of the sequence represented one pixel's spectral information.

The previous RNN-based HSI classifiers have the following demerits: 1) in the image-level method [20], PCA operation destroys the continuity of spectral information; 2) some directional scannings are just summed up directly without considering correlation or weights among them; and 3) one single-sorted pixel sequence cannot fully represent the various and comprehensive spatial dependence.

This study proposes a multiscanning strategy to feature the HSI pixel's sequential specialty and consider the spatial dependence on the pixel level. It is designed for transforming a local HSI patch into several neighboring continuous sequences. Subsequently, considering these scanning directions complement one local spatial patch, we also implement a direction-feature-based scheme to combine complementarity with their correlative weights through an RNN. Based on four stages, the network can automatically implement the end-to-end HSI classification task. Compared with the aforementioned RNN-based methods, the method proposed in this study has three merits: 1) spatial information and spectral information are integrated; 2) correlative dependencies are considered in multiscanning manners; and 3) multiscanning strategy tries to fully compensate for different spatial dependencies.

The main contributions of the proposed network can be described in the following points.

1) The multiscanning strategy using RNNs for extracting spectral–spatial features obtains satisfying performances. Compared with CNN-based methods, the proposed method obtains comparative results with fewer amount of parameters.
2) We investigate the effects of several scanning forms with different scanning orders for the HSI classification.
3) Our method provides a new perspective for RNN-based methods with effective and acceptable scanning directions to convert an image patch into sequential data.
4) The proposed method is a viable alternative to CNNs and provides a perspective for only using RNN to construct spectral information and spatial information acceptably.

The remainder of this article is organized as follows. Section II describes the related works and observations. Section III illustrates our methodology in detail. Section IV focuses on the integrated networks. The experimental results and other analyses are presented in Section V. Finally, in Section VI, we present our conclusions.

II. Related Work and Observations

This section presents three points:1) the meaning of "local" and "global" in patch-based HSI classifiers; 2) the reviews of limited scanning manners for an image into a sequence, and 3) two observations are shown to raise our idea.

A. Description of "Local" and "Global" in Patch-Based Methods

Current deep-learning-based HSI classifiers follow the patch-based learning framework by first dividing the HSI into tons of overlapping patches with patch size w. Next, these patches are denoted as the training set and the testing set. Then, patchwise classification is applied to each patch, and semantic labels obtained by the model are assigned the corresponding central pixels to get the complete classification map [53].

Unlike general image classification tasks, such as ImageNet, which contains a $224 \times 224 \times 3$ RGB image with specific content, in the HSI classification, each "image" is one $w \times w \times C$ patch, where C is the number of channels of the HSI. Hence, concerning each patch, the "global" represents the complete information in this cropped patch instead of the whole original HSI. Furthermore, the "local" represents the small region information in this cropped HSI patch.

To cover the "global" information of the patch, CNNs should be deeper with more feature extraction layers and a large number of parameters [31]. However, RNNs can ensure that each feature activation in its output is an activation at the specific location concerning the "global" patch, in contrast

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to the usual convolution+pooling layer, which only has a "local" context window [32]. Consequently, in RNNs, one shallow feature extraction layer is enough to cover the "global" information of the patch. Our proposed RNN-based method not only enlarges the typical RNN's receptive patterns but also preserves more local continuity information in the cropped HSI patch compared with dilated convolution. Compared with deep CNNs that hierarchically enlarge the receptive field and generate deep features, our proposed method has the ability to generate more discriminative representations from shallow features by a single layer with fewer parameters.

B. Existing Scanning Manner in RNN-Based Methods

In [32], ReNet was proposed to perform object recognition by replacing the convolution+pooling layer with four RNNs that sweep horizontally and vertically in both directions across the image. Zhou *et al.* [20] scanned one channel image row by row in one direction (top to bottom); each row was fed into the RNN as one step. In [29], four directional scanning manners were deployed from left-to-right, right-to-left, top-to-bottom, and bottom-to-top. In [52], four diagonal scannings were introduced on an image region in an acyclic way. Zhang *et al.* [33] scanned one 3-D HSI patch into a pixel sequence by calculating each pixel's Euclidean distance to the central pixel; one step is one pixel with its spectral information. Moreover, in [51], diagonal, vertical, and horizontal scanning manners were deployed on one image patch, and then, the general summation operation from the output of RNNs was performed.

As shown in Fig. 1, one type of method mentioned above aims to consider and combine the possible scanning manners on image level (row by row or column by column) due to their complementarity; however, these methods usually operate on the first PC of the HSI, which ignores the crucial spectral information of the original data. Another method tends to find one optimal sorting manner on pixel level (pixel by pixel); however, one single sorting manner cannot sufficiently find the appropriate representation of the image patch. Moreover, as it is well known, changing the order of input for the RNN will generally output different representations.

In contrast, we aim to fully find the complementary scanning manners on the pixel level and devise one dynamic scheme for weighting the correlative contribution of each scanning manner concerning the image patch.

C. Observations

1) Scanning Manners Make Patterns: We observe that, for one image patch with fixed spatial distribution, different scanning manners will generate different patterns of the pixel sequences, as shown in Fig. 2, which illustrates that there may be various pixel sequences able to represent one patch's spatial information. For instance, the first and second manners start from the top-left pixel and end at the bottom-right pixel with horizontal or vertical scanning manner separately; two types of patterns are generated clearly, but they both serve as candidates. Thus, inputting these two sequences into the RNN will exactly output two different representations for this image patch.

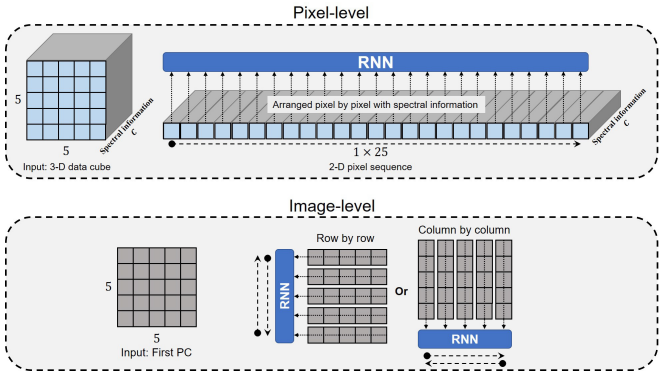


Fig. 1. (Top) Pixel-level conversion; input is original patch, 3-D data cube; one pixel is one step for the RNN. (Bottom) Image-level conversion; input is the first PC of the patch; one row or column is one step for the RNN.

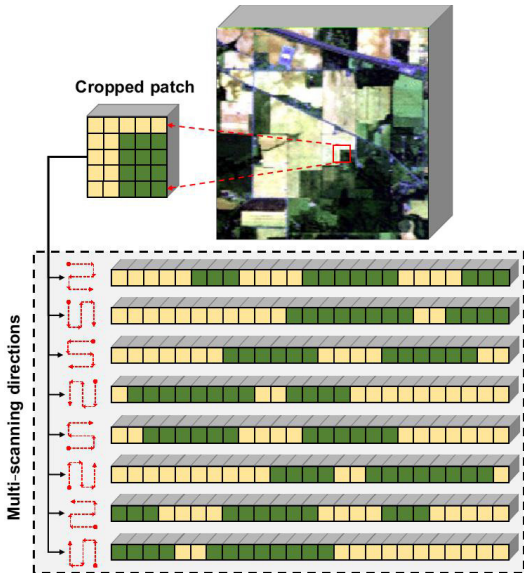


Fig. 2. Observation 1: Different scanning manners on same patch will generate different patterns of pixel sequences.

2) Features Become More Discriminative by Considering More Scanning Sequences: We also investigate how the network seems to get better discriminative power by combining several different pixel sequences randomly. The output representations from the RNN tend to be clearly and discriminatively divided into several clusters by the t-SNE [49] technique, which is also verified by the decreasing total loss of Kullback–Leibler (KL) divergence [50], as shown in Fig. 3. It is clear from Fig. 3(a) and (b) that all clusters distribute scattered and noncollective; several clusters even have a different color. The results become better by increasing the number of pixel sequences; clusters tend to be collective and dense. Therefore, we assume that integrating several pixel sequences is analogous to increase the amount of input information for RNN.

Based on the two observations, we are motivated to generate different scanning pixel sequences from one HSI patch appropriately and comprehensively to fit different spatial dependencies adaptively.

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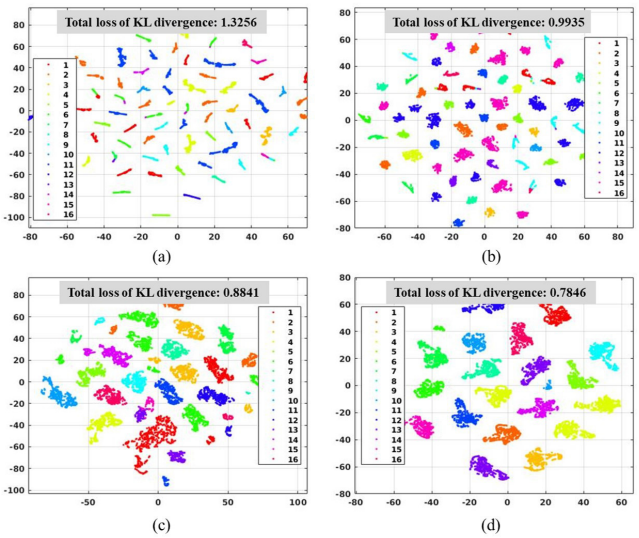


Fig. 3. Observation 2: adding more scanning sequences into the network, the output feature representations can be divided by t-SNE discriminatively. (a) One scanning sequence is used. (b) Two scanning sequences are used. (c) Four scanning sequences are used. (d) Eight scanning sequences are used.

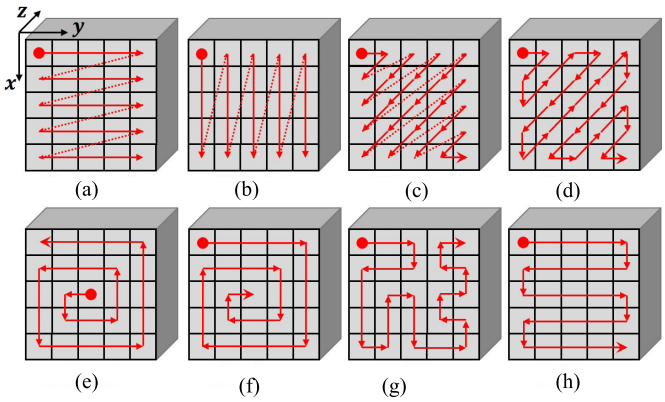


Fig. 4. Investigated scanning forms: (a) horizontal, (b) vertical, (c) diagonal, (d) zig-zag, (e) expansion, (f) perimeter, (g) hilbert, and (h) U-turn. The dashed line is the jump connection.

III. Methodology

This section describes the proposed multiscanning strategy for transforming a local spatial patch of HSI into conceptual and suitable sequential data for RNN.

A. Pixel-Level Scanning Manner

Pixel-level scanning means that one pixel should be one step for the input of the RNN, as shown in Fig. 1(a). Compared with the image-level scanning manner, which needs to implement dimension reduction on original HSI data first, pixel-level scanning can considerably preserve the original spectral information of HSI data. Therefore, it is beneficial to integrate spatial information and spectral information for the HSI classification task.

B. Multiscanning Strategy

1) How to Scan One Image Patch Into Sequential Data?: As we know, one $w \times w$ image patch can be arranged into tons of pixel sequences. Take $w = 5$ as an example; the number of 25 permutations of 25 (P_{25}^{25}) pixel sequences can be generated by changing the order of each pixel possibly. However, most pixel sequences are meaningless and redundant because they fail to catch spatial distribution's spatial structure or continuity. Hence, the restrictions of scanning manners are necessary. Furthermore, several scanning forms are investigated, as shown in Fig. 4

First, the scanned pixel sequences should be spatially continuous to integrate nearest neighbors as much as possible because nonadjacent connections may destroy spatial continuity. Using neighbors' information generally helps to decrease intra-class variance, thereby reducing the pepper-like phenomenon in classification performance [21].

Second, if we convert a local image patch into a sequence, the central step of the sequence can directly depend on the previous step's information and use the following step's information. By changing the perspective, in an image patch, the central pixel can not only directly learn the spatial dependence from the nearest "left" pixel but also "right," "up," and "down" neighboring pixels; patch-based methods aim to classify the central pixel by considering its neighbors. Hence, by scanning one image into a sequence, at least two nearest pixels in the image location should be arranged alongside the corresponding central pixel in the sequence.

Based on the assumptions, Hilbert scanning and U-turn scanning are more appropriate. Take the U-turn scanning form as an example; the scanning strategy is denoted in the following.

One $H \times W \times C$ HSI can be denoted as

$$\boldsymbol{X} = \left\{ \boldsymbol{f}(x, y) \middle| \begin{matrix} y=0,1,2,3,\ldots,W-1 \\ x=0,1,2,3,\ldots,H-1 \end{matrix} \right\} \tag{1}$$

where $\boldsymbol{f}(x, y)$ represents spectral vector (C) at spatial position (x, y). Therefore,

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{f}(0,0) & \boldsymbol{f}(0,1) & \ldots & \boldsymbol{f}(0,W-1) \\ \boldsymbol{f}(1,0) & \boldsymbol{f}(1,1) & \ldots & \boldsymbol{f}(1,W-1) \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{f}(H-1,0) & \boldsymbol{f}(H-1,1) & \ldots & \boldsymbol{f}(H-1,W-1) \end{pmatrix}. \tag{2}$$

In the patch-based methods, tons of patches corresponding to one central pixel will be cropped firstly. For example, one $w \times w \times C$ patch is defined as

$$\boldsymbol{X}_{ij} = \left\{ \boldsymbol{f}(i+\alpha, j+\beta) \middle| \alpha, \beta = -\frac{w-1}{2} \sim \frac{w-1}{2} \right\}. \tag{3}$$

We have to note that $\alpha, \beta \in \mathbb{Z}$. w is the patch size, an odd number. $\boldsymbol{f}(i, j)$ is the central pixel of patch $\boldsymbol{X}_{ij}$. $i + \alpha$ and $j + \beta$ record the position index in $\boldsymbol{X}_{ij}$.

Take $w = 5$ as an example (4), as shown at the bottom of the next page, where $2 \le i \le H - 3$ and $2 \le j \le W - 3$.

Consequently, using the multiscanning strategy, the HSI patch $\boldsymbol{X}_{ij}$ will be transferred into several sequential data $\boldsymbol{d}_{ij}^{s}$ and $s = 1, 2, \ldots, S$. S is the total number of scanning orders under one scanning form.

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TABLE I

INVESTIGATED SCANNING FORMS FOR HORIZONTAL, VERTICAL, DIAGONAL, AND ZIG-ZAG

		Scanning forms							
		Horizontal		Vertical		Diagonal		Zig-zag	
Direction	Order	Start	End	Start	End	Start	End	Start	End
1	$(+x,+y)\rightarrow$	$f(0,0)$	$f(4,4)$	-	-	$f(0,0)$	$f(4,4)$	$f(0,0)$	$f(4,4)$
2	$(+x,+y)\downarrow$	-	-	$f(0,0)$	$f(4,4)$	$f(0,0)$	$f(4,4)$	$f(0,0)$	$f(4,4)$
3	$(-x,-y)\leftarrow$	$f(4,4)$	$f(0,0)$	-	-	$f(4,4)$	$f(0,0)$	$f(4,4)$	$f(0,0)$
4	$(-x,-y)\uparrow$	-	-	$f(4,4)$	$f(0,0)$	$f(4,4)$	$f(0,0)$	$f(4,4)$	$f(0,0)$
5	$(+x,-y)\leftarrow$	$f(0,4)$	$f(4,0)$	-	-	$f(0,4)$	$f(4,0)$	$f(0,4)$	$f(4,0)$
6	$(+x,-y)\downarrow$	-	-	$f(0,4)$	$f(4,0)$	$f(0,4)$	$f(4,0)$	$f(0,4)$	$f(4,0)$
7	$(-x,+y)\rightarrow$	$f(4,0)$	$f(0,4)$	-	-	$f(4,0)$	$f(0,4)$	$f(4,0)$	$f(0,4)$
8	$(-x,+y)\uparrow$	-	-	$f(4,0)$	$f(0,4)$	$f(4,0)$	$f(0,4)$	$f(4,0)$	$f(0,4)$

TABLE II

INVESTIGATED SCANNING FORMS FOR EXPANSION, PERIMETER, HILBERT, AND U-TURN

		Scanning forms							
		Expansion		Perimeter		Hilbert		U-Turn	
Direction	Order	Start	End	Start	End	Start	End	Start	End
1	$(+x,+y)\rightarrow$	$f(2,2)$	$f(4,4)$	$f(0,0)$	$f(2,2)$	$f_1(0,0)f_2(4,0)$	$f_1(0,4)f_2(4,4)$	$f(0,0)$	$f(4,4)$
2	$(+x,+y)\downarrow$	$f(2,2)$	$f(4,4)$	$f(0,0)$	$f(2,2)$	$f_1(0,0)f_2(0,4)$	$f_1(4,0)f_2(4,0)$	$f(0,0)$	$f(4,4)$
3	$(-x,-y)\leftarrow$	$f(2,2)$	$f(0,0)$	$f(4,4)$	$f(2,2)$	-	-	$f(4,4)$	$f(0,0)$
4	$(-x,-y)\uparrow$	$f(2,2)$	$f(0,0)$	$f(4,4)$	$f(2,2)$	-	-	$f(4,4)$	$f(0,0)$
5	$(+x,-y)\leftarrow$	$f(2,2)$	$f(4,0)$	$f(0,4)$	$f(2,2)$	$f(4,4)$	$f(4,0)$	$f(0,4)$	$f(4,0)$
6	$(+x,-y)\downarrow$	$f(2,2)$	$f(4,0)$	$f(0,4)$	$f(2,2)$	$f(0,4)$	$f(0,0)$	$f(0,4)$	$f(4,0)$
7	$(-x,+y)\rightarrow$	$f(2,2)$	$f(0,4)$	$f(4,0)$	$f(2,2)$	$f(4,4)$	$f(0,4)$	$f(4,0)$	$f(0,4)$
8	$(-x,+y)\uparrow$	$f(2,2)$	$f(0,4)$	$f(4,0)$	$f(2,2)$	$f(4,0)$	$f(0,0)$	$f(4,0)$	$f(0,4)$

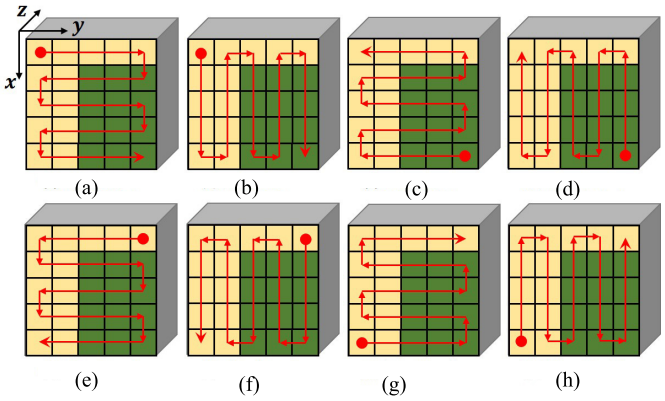


Fig. 5. Eight scanning orders of U-turn form on one 5×5 image patch with fixed spatial distribution. (a) Direction-1. (b) Direction-2. (c) Direction-3. (d) Direction-4. (e) Direction-5. (f) Direction-6. (g) Direction-7. (h) Direction-8.

If $i, j = 2$, take U-turn scanning direction-1 $\boldsymbol{d}_{22}^{1}$ with patch size $w = 5$ as an example shown in Fig. 5(a); it can be ordered as

$$\begin{aligned} &f(0,0)\rightarrow f(0,1)\rightarrow f(0,2)\rightarrow f(0,3)\rightarrow f(0,4)\\ &\quad\rightarrow f(1,4)\rightarrow f(1,3)\rightarrow f(1,2)\rightarrow f(1,1)\rightarrow f(1,0)\\ &\quad\rightarrow f(2,0)\rightarrow f(2,1)\rightarrow f(2,2)\rightarrow f(2,3)\rightarrow f(2,4)\\ &\quad\rightarrow f(3,4)\rightarrow f(3,3)\rightarrow f(3,2)\rightarrow f(3,1)\rightarrow f(3,0)\\ &\quad\rightarrow f(4,0)\rightarrow f(4,1)\rightarrow f(4,2)\rightarrow f(4,3)\rightarrow f(4,4). \end{aligned} \tag{5}$$

Therefore, the pixel sequence $\boldsymbol{d}_{22}^{1}$ can be generated as

$$\begin{aligned} \boldsymbol{d}_{22}^{1} = \big[&f(0,0), f(0,1), f(0,2), f(0,3),\\ &f(0,4), f(1,4), \ldots, f(4,4)\big]. \end{aligned} \tag{6}$$

Other investigated scanning directions under the same conditions ($i, j = 2$ and $w = 5$) are shown in Tables I and II. In the tables, each scanning form with different scanning orders is illustrated by the "Start" pixel and "End" pixel. "-" represents the noninvestigated ordering situation. $\uparrow, \downarrow, \leftarrow, \rightarrow$ represent the beginning direction of scanning orders. It is better to check the U-turn scanning form in Table II with Fig. 5.

2) How Many Scanned Pixel Sequences Should Be Used?: Depending on the previous assumptions, eight acceptable U-turn scanning orders are generated to consider the effect of each neighborhood pixel on the central pixel. They can be seen as the comprehensive scanning directions from top-left, bottom-right, top-right, and bottom-right pixels separately on original HSI patches.

This transform's specialty is that, no matter how we convert the patch into a sequence with different directions, the central pixel always stays in the middle of the pixel sequence.

$$\boldsymbol{X}_{ij} = \begin{pmatrix} f(i-2,j-2) & f(i-2,j-1) & f(i-2,j) & f(i-2,j+1) & f(i-2,j+2)\\ f(i-1,j-2) & f(i-1,j-1) & f(i-1,j) & f(i-1,j+1) & f(i-1,j+2)\\ f(i,j-2) & f(i,j-1) & f(i,j) & f(i,j+1) & f(i,j+2)\\ f(i+1,j-2) & f(i+1,j-1) & f(i+1,j) & f(i+1,j+1) & f(i+1,j+2)\\ f(i+2,j-2) & f(i+2,j-1) & f(i+2,j) & f(i+2,j+1) & f(i+2,j+2) \end{pmatrix} \tag{4}$$

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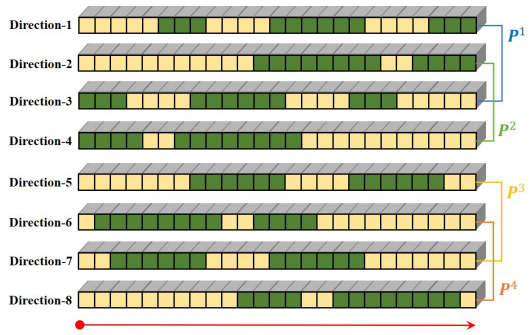


Fig. 6. Four pairs of flattened forward and backward pixel sequences from U-turn scanning form. P^i connects two totally reversed pixel sequences.

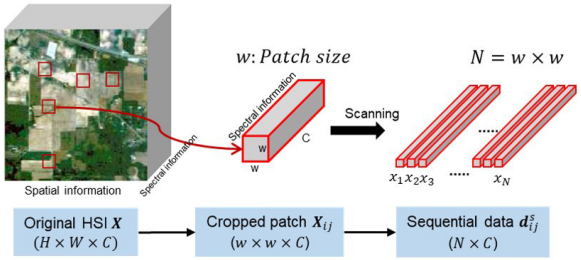


Fig. 7. Spectral–spatial structure for the patch of HSI.

Hence, it can learn reasonable spatial correlations from different directions with the same time steps.

It is clear from Fig. 6 that, by scanning one image patch into various pixel sequences, the distribution of their patterns is different, which means that, for one image patch with fixed spatial distribution, there may be various possible ways to represent this image patch. In the CNN-based methods, only one image patch is fed as an input; the information is fixed and single. However, by scanning one image patch into several pixel sequences, these sequences can be fed into RNN models. In other words, the multiscanning strategy enlarges the number of data in terms of the sequential data and increases the amount of input information.

Moreover, in the scanned pixel sequences, we find that each is paired with another one, such as direction-1 and direction-4. As shown in Fig. 6, they are reversed. The forward and backward direction sequences are better for constructing a bidirectional RNN (Bi-RNN), which can get more comprehensive reception steps and consider the whole sequence. Hence, we group the paired sequences and get four pairs, P^1, P^2, P^3, and P^4. Its detail will be illustrated in Section IV.

C. Spectral–Spatial Structure of HSI for RNN

HSI experiments are divided into spatial and spectral domains in most existing methods, requiring practical preprocessing skills, and are time-consuming. We unify the spatial domain and the spectral domain together using the RNN and multiscanning strategy, as briefly shown in Fig. 7.

Take one $H \times W \times C$ HSI image as an example: first, the cropped local patch $\boldsymbol{X}_{ij}$ with patch size w, a 3-D data cube $(w \times w \times C)$, will be transferred into several sequential data (pixel sequence) $\boldsymbol{d}^s_{ij}$ by multiscanning strategy.

Each element is a pixel vector containing the spectral information of one pixel with dimension C in the original data in each pixel sequence. Thus, each element will be a step for the RNN, and the number of steps should equal the number of pixels $N = w \times w$ in this patch.

The spectral–spatial structure is converted into a neighboring relevant sequence. In this way, spectral information and spatial information are integrated simultaneously, and the relationship among the neighboring local pixels is considered effective. Hence, the proposed spectral–spatial structure of HSI in the concept and structure is appropriate as the input of the RNN model.

D. RNN for Multiscanning Sequences

RNN is one of the augmented feed-forward neural networks [22] as it constructs the connections between adjacent time steps [23]. Unlike the general feed-forward neural network, the RNN model can learn the characteristics or patterns in sequential data. In each step of the sequential data, the current activation depends on the previous step's activation. Therefore, the last step's output has the "memory," which influences the network's final decision [24].

Introducing the RNN into our multiscanning strategy, for each scanned pixel sequence $\boldsymbol{d}^s_{ij}$, the hidden state $\boldsymbol{h}^s_{ij(n)}$ at the nth-order pixel can be updated by the previous $n-1$th hidden state $\boldsymbol{h}^s_{ij(n-1)}$ with current nth input $\boldsymbol{d}^s_{ij}[n]$ through

$$\boldsymbol{h}^s_{ij(n)} = \begin{cases} 0, & \text{if } n = 0 \\ \boldsymbol{\rho}_h\left(\boldsymbol{h}^s_{ij(n-1)}, \boldsymbol{d}^s_{ij}[n]\right), & \text{otherwise} \end{cases} \tag{7}$$

where $0 \le n \le N-1$ and $N = w \times w$ with patch size w. $\boldsymbol{\rho}_h$ is a nonlinear function, such as a logistic sigmoid function or a hyperbolic tangent function. The update rule of the recurrent hidden state in (7) is usually implemented as follows:

$$\boldsymbol{h}^s_{ij(n)} = \boldsymbol{\rho}_h\left(\boldsymbol{W}^s_{ij}\boldsymbol{d}^s_{ij}[n] + \boldsymbol{U}^s_{ij}\boldsymbol{h}^s_{ij(n-1)} + \boldsymbol{b}^s_{ij}\right) \tag{8}$$

where $\boldsymbol{W}^s_{ij}$ represents the input-to-hidden weight matrix, $\boldsymbol{U}^s_{ij}$ is the hidden-to-hidden weight matrix between the hidden layer of two adjacent pixel orders, and $\boldsymbol{b}^s_{ij}$ is the bias, in scanning order s of pixel sequence $\boldsymbol{d}^s_{ij}$.

Subsequently, the output $\boldsymbol{y}^s_{ij}$ from last hidden state $\boldsymbol{h}^s_{ij(N-1)}$ at scanning order s is calculated by

$$\boldsymbol{y}^s_{ij} = \boldsymbol{\rho}_y\left(\boldsymbol{Z}^s_{ij}\boldsymbol{h}^s_{ij(N-1)} + \boldsymbol{b'}^s_{ij}\right) \tag{9}$$

where $\boldsymbol{Z}^s_{ij}$ represents the hidden-to-output weight matrix and $\boldsymbol{b'}^s_{ij}$ is the bias.

Hence, in the multiscanning strategy, several flattened HSI pixel sequence $\boldsymbol{d}^s_{ij}$ will be the input for the RNN model separately. The last step output $\boldsymbol{y}^s_{ij}$ of each scanned pixel sequence will be used as output representations for each scanning order s. We note it as $\boldsymbol{F}^s_{ij}$. Therefore,

$$\boldsymbol{F}_{ij} = \left\{\boldsymbol{F}^s_{ij} \middle| s = 1, 2, 3, \ldots, S\right\} \tag{10}$$

where S is total number of scanning orders in a scanning form. Hence, $\boldsymbol{F}_{ij}$ is the total generated feature set.

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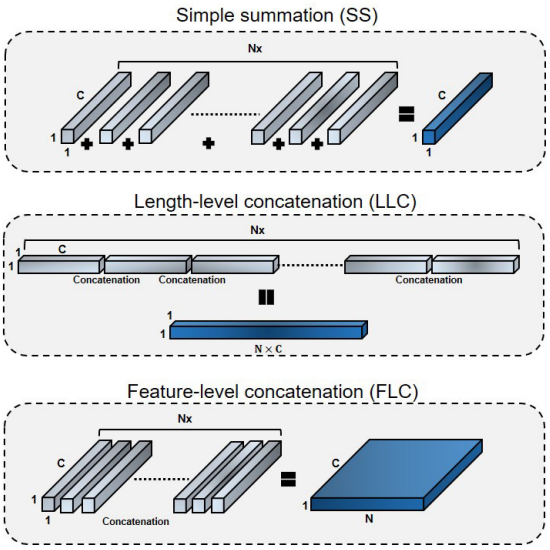


Fig. 8. Possible combination manners. Gray cube represents output feature from RNN of each scanning order. Blue cube represents the combined feature.

In the experiments, the general RNN (RNN), gated recurrent unit (GRU) [27] and long short-term memory (LSTM) [26] will be investigated for classification performance.

E. Weighted Feature Concatenation

As described before, in the multiscanning strategy, each scanning order complements one image patch. To fully take advantage of this complementary information, an appropriate combination is necessary. Fig. 8 shows the possible combination manners. Moreover, we will investigate and experiment with them separately later.

Typically, a simple summation or average of each scanning feature is one of the direct ways to combine them. However, it means that each contributes equally to the classification task even for different spatial dependencies. The fact is that the multiscanning strategy deals with different situations of spatial dependence in the HSI patch; the spatial dependence from different scanning orders is changeable. Therefore, some scanning directions should have a smaller weight, while those key scanning directions should have a greater weight.

Motivated by [14] and [45] introducing the attention mechanism, an intuitive method is to construct relations between these scanning orders. Hence, we attempt to combine all the output features from the RNN in a weighted manner. Specifically, the weighted feature can be formulated as (11) and (12)

$$\boldsymbol{e}_{ij}^{s} = \mathrm{ReLU}\left(\boldsymbol{W}_{e}^{s} \odot \boldsymbol{F}_{ij}^{s} + \boldsymbol{b}_{e}^{s}\right). \tag{11}$$

Equation (11) is a single-layer neural network; $\odot$ denotes the dot product. This layer rearranges the output of each RNN in its current vector space, and then, the ReLU activation function transforms it to get $\boldsymbol{e}_{ij}^{s}$ as a new feature representation of $\boldsymbol{F}_{ij}^{s}$. $\boldsymbol{W}_{e}^{s}$ is the transformation matrix. $\boldsymbol{b}_{e}^{s}$ is the bias. The attention weight w_{ij}^{s} is produced through the Softmax layer, formulated as follows:

$$w_{ij}^{s} = \mathrm{Softmax}\left(\boldsymbol{W}_{w}^{s} \odot \boldsymbol{e}_{ij}^{s} + b_{w}^{s}\right) \tag{12}$$

where $w_{ij}^{s} \in \mathbb{R}^{1}$ and $s \in 1, 2, 3, \ldots, S$. We measure the importance based on the relevance of $\boldsymbol{e}_{ij}^{s}$ with other channelwise vectors. $\boldsymbol{W}_{w}^{s}$ is the transformation matrices, b_{w}^{s} is the bias term, and the Softmax maps the nonnormalized output to a probability distribution and constrain the output to be in the interval (0, 1).

After obtaining the newly learned attention weights, we update the feature representation $\boldsymbol{F}_{ij}^{s}$ at each scanning order s using

$$\tilde{\boldsymbol{F}}_{ij}^{s} = w_{ij}^{s} \times \boldsymbol{F}_{ij}^{s}. \tag{13}$$

These weights can be integrated into the entire network, and their optimal values are automatically learned from data. Therefore, the output features from the multiscanning strategy can be renoted as

$$\tilde{\boldsymbol{F}}_{ij} = \{\tilde{\boldsymbol{F}}_{ij}^{s} | s = 1, 2, 3, \ldots, S\}. \tag{14}$$

With the allocated weights for the different scanning orders, our model can obtain a more reasonable explanation that some scanning orders play a key role, and some of them are less important. Meanwhile, our model can better characterize the spectral–spatial features of the HSI patches, find the correlations of neighboring pixels, and pay more attention to the appropriate transformation of HSI patches, which may make the training model more accurate.

IV. Integrated Network

This section describes the integrated network structures. Two schemes are implemented as shown in Figs. 9 and 10. In the first scheme, each direction sequence is fed into a single LSTM separately. The second scheme integrates the bidirectional LSTM (Bi-LSTM) for four pairs of forwarding and backward pixel sequences. Subsequently, learned features with direction-based attentions would be concatenated for complementarity for one HSI patch.

A. Scheme-1

In this scheme, multiscanning direction sequences are fed into a single LSTM unit separately, and then, each output representation is concatenated again for capturing the complementarity, as shown in Fig. 9.

First, one cropped HSI patch will be flattened into several pixel sequences by a multiscanning strategy. Second, each directional pixel sequence is fed into the LSTM unit separately. The feature representation $\boldsymbol{F}_{ij}^{s}$ represents each scanned pixel sequence. This feature contains the spatial dependence among all pixels in the patch at scanning order s. Third, learned features of all pixel sequences are combined with their learnable fusion weights together by the feature-level concatenation as

$$\tilde{\boldsymbol{F}}_{ij}^{C} = \mathrm{concat}\left[\tilde{\boldsymbol{F}}_{ij}^{1}, \tilde{\boldsymbol{F}}_{ij}^{2}, \tilde{\boldsymbol{F}}_{ij}^{3}, \ldots, \tilde{\boldsymbol{F}}_{ij}^{s}\right], \quad s = 1, 2, 3, \ldots, S \tag{15}$$

where $\tilde{\boldsymbol{F}}_{ij}^{C}$ is a new sequential data.

In the real experiments, S is set up as 8. Finally, the concatenated feature $\tilde{\boldsymbol{F}}_{ij}^{C}$ is input into the next LSTM to capture the correlative dependence and complementarity between these

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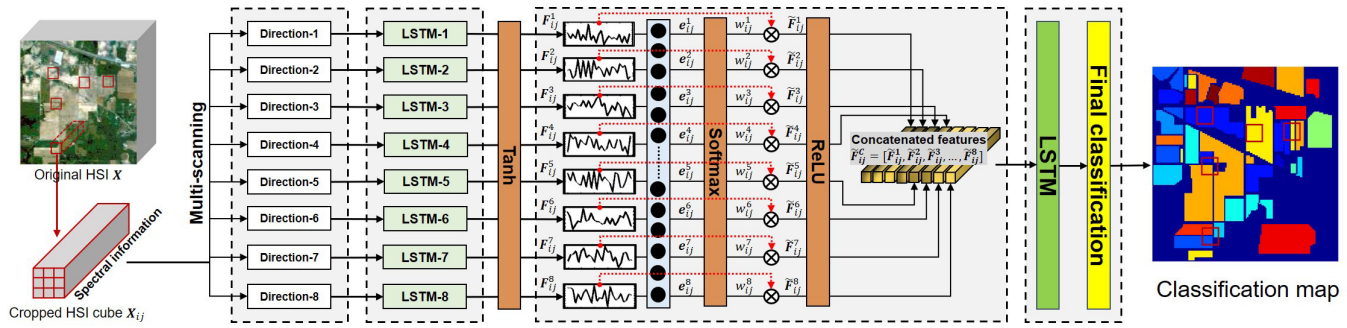


Fig. 9. Scheme-1: multiscanning-based model with separate procedure of each direction. F^i is the output feature from the last step of each LSTM.

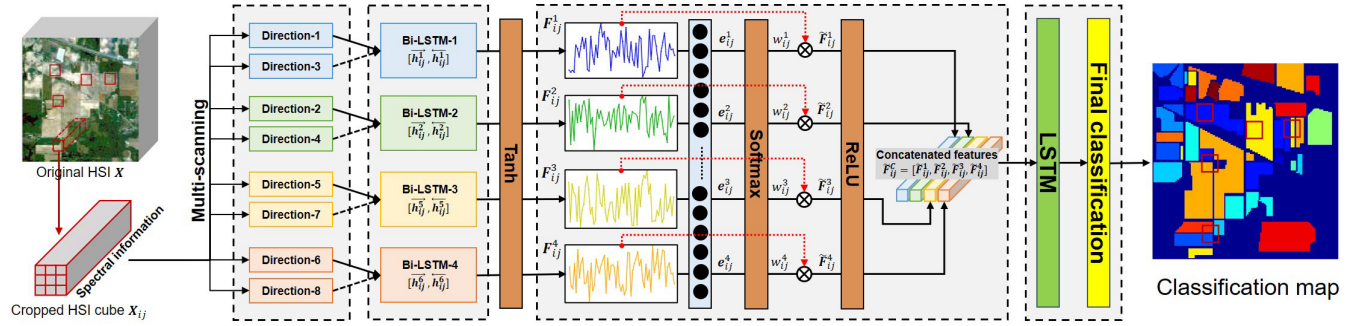


Fig. 10. Scheme-2: the framework with Bi-LSTM for the pairs of scanning directions. F^i is the output feature from each Bi-LSTM.

scanning orders as following procedure: $\tilde{\boldsymbol{F}}_{ij}^1 \rightarrow \tilde{\boldsymbol{F}}_{ij}^2 \rightarrow \tilde{\boldsymbol{F}}_{ij}^3 \rightarrow \tilde{\boldsymbol{F}}_{ij}^4 \rightarrow \tilde{\boldsymbol{F}}_{ij}^5 \rightarrow \tilde{\boldsymbol{F}}_{ij}^6 \rightarrow \tilde{\boldsymbol{F}}_{ij}^7 \rightarrow \tilde{\boldsymbol{F}}_{ij}^8$, therefore referring to (7) and (8)

$$\boldsymbol{h}_{ij(s)}^C = \rho_h\left(\boldsymbol{W}_{ij}^C \tilde{\boldsymbol{F}}_{ij}^C[s] + \boldsymbol{U}_{ij}^C \boldsymbol{h}_{ij(s-1)}^C + \boldsymbol{b}_{ij}^C\right) \tag{16}$$

where $\boldsymbol{h}_{ij(s)}^C$ represents the hidden state at sth scanning order with sth input $\tilde{\boldsymbol{F}}_{ij}^C[s]$. Here, $\tilde{\boldsymbol{F}}_{ij}^C[1] = \tilde{\boldsymbol{F}}_{ij}^1$. $\boldsymbol{W}_{ij}^C$ is the input-to-hidden weight matrix. $\boldsymbol{U}_{ij}^C$ is the hidden-to-hidden weight matrix. $\boldsymbol{b}_{ij}^C$ is the bias.

The output representation of the last scanning order S will be used for classification

$$\boldsymbol{y}_{ij(S)}^C = \boldsymbol{\rho}_y(\boldsymbol{Z}_{ij}^C \boldsymbol{h}_{ij(S)}^C + \boldsymbol{b'}_{ij}^C) \tag{17}$$

where $\boldsymbol{W}_{ij}^C$ represents the weight matrix and $\boldsymbol{b'}_{ij}^C$ is the bias. $\boldsymbol{\rho}_y$ is the activation function.

Therefore, for the input image patch $\boldsymbol{X}_{ij}$, we will get one final feature representation $\boldsymbol{y}_{ij(S)}^C$.

B. Scheme-2

As mentioned before, in the multiscanning strategy, one scanning form can generate four pairs of forwarding and inverse scanning orders, and it is intuitive that they can be fed into Bi-RNN to obtain wider reception fields and make our model more concise and effective; the Scheme-2 model is shown in Fig. 10.

RNNs have achieved remarkable performance in dealing with sequential data analysis [25]. Unlike the general RNN, Bi-RNN can integrate two hidden layers to simultaneously process two opposite directions to generate a joint output, allowing considering information from both forward and backward states. By processing both input data directions, the model has a wider receptive field. The calculation processes in the Bi-RNN are given as follows:

$$\overrightarrow{\boldsymbol{h}_{ij(n)}^s} = \rho_h\left(\overrightarrow{\boldsymbol{W}_{ij}^s} \boldsymbol{d}_{ij}^s[n] + \overrightarrow{\boldsymbol{U}_{ij}^s} \boldsymbol{h}_{ij(n-1)}^s + \overrightarrow{\boldsymbol{b}_{ij}^s}\right) \tag{18}$$

$$\overleftarrow{\boldsymbol{h}_{ij(n)}^s} = \rho_h\left(\overleftarrow{\boldsymbol{W}_{ij}^s} \boldsymbol{d}_{ij}^s[n] + \overleftarrow{\boldsymbol{U}_{ij}^s} \boldsymbol{h}_{ij(n+1)}^s + \overleftarrow{\boldsymbol{b}_{ij}^s}\right) \tag{19}$$

where $0 \le n \le N-1$ and N is the total number of pixels in the patch. $\overrightarrow{\boldsymbol{h}_{ij(n)}^s}$ and $\overleftarrow{\boldsymbol{h}_{ij(n)}^s}$ represent the forward and backward hidden states at the nth-order pixel in pixel sequence $\boldsymbol{d}_{ij}^s$; weights $\overrightarrow{\boldsymbol{W}_{ij}^s}$ and $\overleftarrow{\boldsymbol{W}_{ij}^s}$ represent the input-to-hidden coefficient matrices in the forward and backward manners, respectively. $\overrightarrow{\boldsymbol{U}_{ij}^s}$ and $\overleftarrow{\boldsymbol{U}_{ij}^s}$ denote the hidden-to-hidden coefficient matrices in the forward and backward, respectively. $\overrightarrow{\boldsymbol{b}_{ij}^s}$ and $\overleftarrow{\boldsymbol{b}_{ij}^s}$ are biases in the inverse manner.

Next, the hidden state at the nth order, $\boldsymbol{h}_{ij(n)}^s$, is obtained by concatenating forward and backward hidden states $\overrightarrow{\boldsymbol{h}_{ij(n)}^s}$ and $\overleftarrow{\boldsymbol{h}_{ij(n)}^s}$

$$\boldsymbol{h}_{ij(n)}^s = \text{concat}\left[\overrightarrow{\boldsymbol{h}_{ij(n)}^s}, \overleftarrow{\boldsymbol{h}_{ij(n)}^s}\right]. \tag{20}$$

The output $\boldsymbol{y}_{ij}^s$ from Bi-RNN of two inverse sequences can be referred to (9) at the $(N-1/2)$th step

$$\boldsymbol{h}_{ij(\frac{N-1}{2})}^s = \text{concat}\left[\overrightarrow{\boldsymbol{h}_{ij(\frac{N-1}{2})}^s}, \overleftarrow{\boldsymbol{h}_{ij(\frac{N-1}{2})}^s}\right]. \tag{21}$$

$$\boldsymbol{y}_{ij}^s = \boldsymbol{\rho}_y\left(\boldsymbol{Z}_{ij}^s \boldsymbol{h}_{ij(\frac{N-1}{2})}^s + \boldsymbol{b'}_{ij}^s\right) \tag{22}$$

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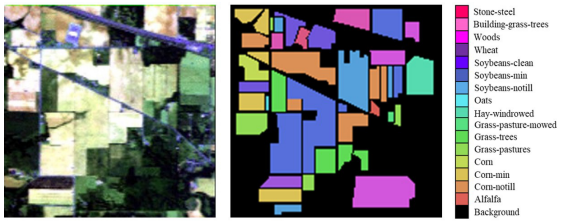


Fig. 11. Indian Pines dataset false color composite image and corresponding ground-truth areas representing 16 land-cover classes.

where $\boldsymbol{Z}_{ij}^{s}$ represents the hidden-to-output weight matrix and $\boldsymbol{b'}_{ij}^{s}$ is the bias. We have to note that, in this place, $s = 1, 2, 3, 4$.

Hence, four pairs of pixel sequences are fed into Bi-RNN. Subsequently, the learned feature representation $\boldsymbol{y}_{ij}^{1}, \boldsymbol{y}_{ij}^{2}, \boldsymbol{y}_{ij}^{3}, \boldsymbol{y}_{ij}^{4}$ of each pair goes through the tanh activation function to obtain $\boldsymbol{F}_{ij}^{1}, \boldsymbol{F}_{ij}^{2}, \boldsymbol{F}_{ij}^{3}$, and $\boldsymbol{F}_{ij}^{4}$. Next, these features are combined together with their learnable fusion weights $w_{ij}^{1}, w_{ij}^{2}, w_{ij}^{3}$, and w_{ij}^{4} as

$$\tilde{\boldsymbol{F}}_{ij}^{s} = w_{ij}^{s} \times \boldsymbol{F}_{ij}^{s}, \quad s = 1, 2, 3, 4. \tag{23}$$

$$\tilde{\boldsymbol{F}}_{ij}^{C} = \text{concat}[\tilde{\boldsymbol{F}}_{ij}^{1}, \tilde{\boldsymbol{F}}_{ij}^{2}, \tilde{\boldsymbol{F}}_{ij}^{3}, \tilde{\boldsymbol{F}}_{ij}^{4}]. \tag{24}$$

Meanwhile, each $\tilde{\boldsymbol{F}}_{ij}^{s}$ is activated by the ReLU function, which is good at building a sparse representation feature because, in terms of representation learning, the sparse representations seem to be better and effective than the dense representations [34].

Finally, the concatenated feature $\tilde{\boldsymbol{F}}_{ij}^{C}$ is input into next LSTM to capture the complementarity and correlation between these scanning orders as follows: $\tilde{\boldsymbol{F}}_{ij}^{1} \rightarrow \tilde{\boldsymbol{F}}_{ij}^{2} \rightarrow \tilde{\boldsymbol{F}}_{ij}^{3} \rightarrow \tilde{\boldsymbol{F}}_{ij}^{4}$.

V. Experimental Results

A. Description of Datasets

The Indian Pines (IP) dataset refers to a mixed vegetation site. This dataset has 145 × 145 pixels and 220 spectral bands. After removing water absorption bands, the number of the bands is reduced to 200. Thus, there are 16 different land-cover classes in the ground truth. The false-color composite image and the ground-truth maps are shown in Fig. 11.

The Pavia University (PU) dataset has 115 bands with 610 × 340 pixels. In the experiment, 12 noisy channels are removed, and the remaining 103 bands are used for the classification task. The original image and ground truth are shown in Fig. 12. It includes nine ground cover types.

The Salinas (SA) dataset comprises 224 bands, and it has a size of 512 × 217 pixels. As with the Indian Pines scene, we discarded the 20 water absorption bands, and 204 bands are left. The false-color image and 16 classes of the ground truth are shown in Fig. 13.

B. Experiment Design

Our proposed method is compared with several well-known methods, including spectral-based methods, spectral CNN [12] and deep RNN [13], spatial-based method, FCN [18], spectral–spatial-based method, 3-D CNN [21], M3D-DCNN [35], and SSAN [39]. It should be noted that the parameter setting of these compared methods comes from their references.

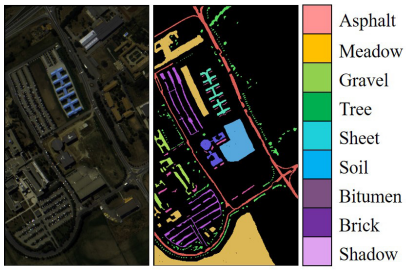


Fig. 12. University of Pavia dataset, false color composite image, and the corresponding ground-truth areas representing nine land-cover classes.

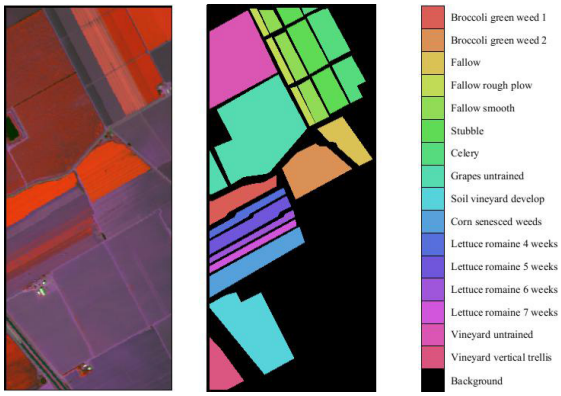


Fig. 13. Salinas dataset, false color composite image, and the corresponding ground-truth areas representing 16 land-cover classes.

In training, the batch size is set as 64. For a relatively fair comparison, all the methods are trained with the Adam algorithm with an initial learning rate of 0.01. Moreover, the number of training epochs is set as 200. Finally, we introduce the dropout layer with a probability of 0.5 and batch normalization layer before the fully connected layer for training the valid supervised model. For reproducibility, the experimental details of our method are written in the following. All the LSTM layer is set as 1; the number of hidden units in LSTM is 64. For the Bi-LSTM, two layers are set: one for the forward direction and another for the backward direction; the number of hidden units for Bi-LSTM is also set as 64. Moreover, all the experiments are implemented on Pytorch 1.1 platform using Geforce RTX 1080 GPU.

Labeled samples are divided into training and testing sets to evaluate the proposed methods' performance and feasibility. We set 10% pixels as training samples in all methods, including compared methods, and the remaining 90% pixels are used for testing. The numbers of the sample selection process are shown in Tables III–V.

The classification performance is evaluated by overall accuracy (OA), average accuracy (AA), kappa coefficient (kappa), and accuracy in each class.

C. Experimental Results

1) Indian Pines Dataset: Fig. 14 shows the classification maps with different methods on the Indian Pines dataset. Quantitative results for each method are also listed in Table VI. First, for the proposed schemes, the patch size is set as 5 for

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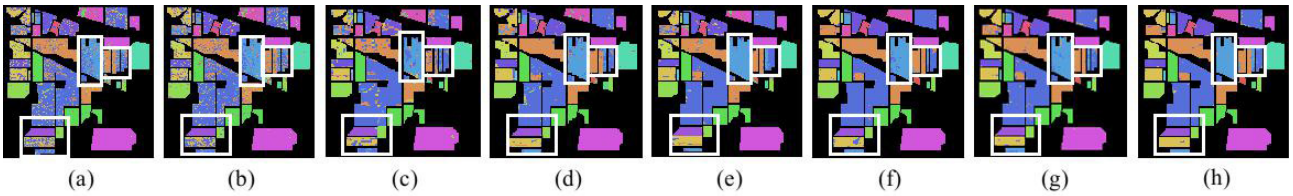


Fig. 14. Classification maps with different methods for the Indian Pines image. (a) Spectral CNN. (b) Deep RNN. (c) FCN. (d) 3-D CNN. (e) M3D-DCNN. (f) SSAN. (g) Proposed Scheme-1. (h) Proposed Scheme-2.

TABLE III
LAND-COVER TYPES AND THE NUMBER OF PIXELS ON THE INDIAN PINES DATASET

Class		No. of Samples	
Type No.	Name	Train	Test
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture	48	435
6	Grass-trees	73	657
7	Grass-pasture-mowed	3	25
8	Hay-windowed	48	430
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	246	2209
12	Soybean-clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Building-grass-trees	39	349
16	Stone-steal-towers	9	84
Total		1025	9224

TABLE IV
LAND-COVER TYPES AND THE NUMBER OF PIXELS ON THE PAVIA UNIVERSITY DATASET

Class		No. of Samples	
Type No.	Name	Train	Test
1	Asphalt	663	5967
2	Meadows	1864	16784
3	Gravel	209	1889
4	Trees	306	2757
5	Metal sheets	134	1210
6	Bare soil	502	4526
7	Bitumen	133	1197
8	Bricks	368	3313
9	Shadow	94	852
Total		3860	38924

TABLE V
LAND-COVER TYPES AND THE NUMBER OF PIXELS ON THE SALINAS DATASET

Class		No. of Samples	
Type No.	Name	Train	Test
1	Broccoli green weed 1	200	1808
2	Broccoli green weed 2	372	3353
3	Fallow	197	1778
4	Fallow rough plow	139	1254
5	Fallow smooth	268	2410
6	Stubble	395	3563
7	Celery	357	3221
8	Grapes untrained	1127	10143
9	Soil vineyard develop	620	5582
10	Corn senesced weeds	327	2950
11	Lettuce romaine 4 weeks	106	961
12	Lettuce romaine 5 weeks	192	1734
13	Lettuce romaine 6 weeks	91	824
14	Lettuce romaine 7 weeks	107	963
15	Vineyard untrained	726	6541
16	Vineyard vertical trellis	180	1626
Total		11219	48711

competitive comparison although some other methods set different patch sizes. To obtain fair comparison results, we changed their patch size to 5. The internal structures and settings follow the references.

As shown in Fig. 14 and Table VI, the spectral-based methods, spectral CNN, and deep RNN only use one pixel's spectral information without considering its local spatial (neighborhood) pixels. As a result, the classification results of these two methods are very poor. The noise-like misclassifications are obvious, and it appears scattered severely. In the HSI, it is highly possible for the spatially nearby pixels to share the same class label. For example, in Fig. 14, several bigger and flatter regions are marked by white rectangular. In these regions, the pixels ought to be classified into the same label. However, due to the fact that different materials may have the same spectral signatures and the same material may possess different variations, the misclassifications are unordered, irregular, and ubiquitous. In Table VI, spectral-based methods present some unsatisfying results, such as class "4," class "9," and class "15." Most of them result from unbalanced training samples and the complex situation in spectral signatures.

Although FCN takes account of the spatial information into the training procedure, preprocessing for dimension reduction is deployed, which destroys the band-to-band correlation in the spectral domain. In Fig. 14(c), we can clearly find relatively fewer noise-like misclassifications, but there exist object-like misclassifications, which means that the FCN actually considered the spatial pixels' correlations. If we look closely at white boxes, we will find that object-like misclassifications vary and primarily appear on class regions' outlines.

It is informative and intuitive to combine spectral information and spatial information. Therefore, 3-D CNN is suitable for its structure. 3-D CNN, M3D-DCNN, and SSAN achieved better results due to the consideration between spectral information and spatial information. In detail, Fig. 14(d)–(f) shows the classification results by 3D-CNN, M3D-DCNN, and SSAN separately. It is clear that, by exploring the spectral information and spatial information, the results become satisfying. In the big regions marked by the white box, the classification results are flatter and smoother. However, in the small-sized HSI, one patch may be heterogeneous, which means that 3-D convolution may convolve different variations into one position of the feature map, affecting the final classification results. For example, we can clearly find that some class regions'

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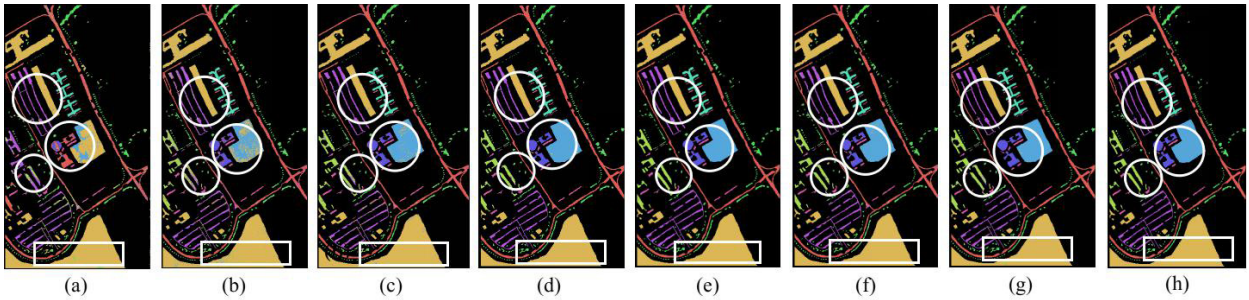


Fig. 15. Classification maps with different methods for the Pavia University image. (a) Spectral CNN. (b) Deep RNN. (c) FCN. (d) 3-D CNN. (e) M3D-DCNN. (f)SSAN. (g) Proposed Scheme-1. (h) Proposed Scheme-2.

TABLE VI
CLASSIFICATION ACCURACY (%) FOR THE INDIAN PINES IMAGE ON THE TEST SET

Classes	Methods							
	spectral CNN	Deep-RNN	FCN	3-D CNN	M3D-DCNN	SSAN	Scheme-1(ours)	Scheme-2(ours)
1	21.78	76.71	40.82	80.05	63.33	**88.68**	86.66	87.27
2	62.77	74.90	75.31	82.57	88.75	93.34	93.58	**95.68**
3	63.31	61.44	57.88	72.19	81.16	90.75	92.17	**94.75**
4	54.94	56.30	61.29	87.23	84.09	84.37	90.33	**91.77**
5	36.80	92.99	91.62	93.72	93.28	97.21	96.99	**98.58**
6	86.33	94.72	95.38	97.81	97.79	**98.67**	97.36	98.55
7	80.02	92.68	60.66	**95.48**	81.19	86.20	88.91	91.51
8	93.05	97.95	95.16	98.41	98.41	**100**	**100**	**100**
9	0	60.05	56.02	60.88	86.77	78.89	94.16	**100**
10	52.71	70.13	67.68	75.67	90.30	**95.33**	93.02	95.30
11	70.35	75.06	75.33	83.42	91.64	93.85	94.11	**97.05**
12	57.79	76.89	73.79	80.57	82.07	95.52	95.31	**97.38**
13	88.52	98.81	98.88	99.66	**100**	**100**	**100**	**100**
14	84.36	94.37	93.32	96.49	97.66	97.40	96.18	**98.91**
15	43.69	71.89	68.37	75.23	68.25	**82.29**	79.58	80.33
16	91.44	92.23	88.59	97.88	93.88	**99.03**	98.37	98.34
OA	69.80	79.94	79.12	85.49	90.22	**95.57**	95.13	95.38
AA	61.74	80.45	75.01	86.08	88.41	94.26	93.55	**95.45**
Kappa	0.64	0.74	0.75	0.85	0.88	**0.95**	0.93	**0.95**

TABLE VII
CLASSIFICATION ACCURACY (%) FOR THE PAVIA UNIVERSITY IMAGE ON THE TEST SET

Classes	Methods							
	spectral CNN	Deep-RNN	FCN	3-D-CNN	M3D-DCNN	SSAN	Scheme-1(ours)	Scheme-2(ours)
1	92.60	92.23	96.64	96.05	98.61	98.65	99.19	**99.88**
2	92.91	95.91	96.24	97.99	98.10	**98.83**	97.29	97.97
3	76.01	70.89	92.68	88.98	90.19	93.65	94.11	**95.25**
4	92.00	95.33	97.15	97.35	99.45	98.09	99.31	**99.96**
5	99.33	99.91	99.91	**100**	**100**	**100**	**100**	**100**
6	77.28	90.19	97.08	96.12	98.29	98.93	99.02	**99.48**
7	85.88	79.87	95.72	89.68	98.44	96.75	96.71	**98.46**
8	84.52	80.88	94.95	90.65	95.88	**98.34**	98.04	98.29
9	99.88	99.88	99.88	99.88	**100**	98.55	99.33	**100**
OA	90.02	91.99	95.45	96.36	98.07	99.02	98.99	**99.18**
AA	88.97	89.45	96.69	95.19	97.66	97.85	98.11	**98.81**
Kappa	0.83	0.86	0.95	0.95	0.97	**0.98**	**0.98**	**0.98**

boundaries are unclear, and the misclassifications classes occur between some adjacent class regions. SSAN considers the model's spatial attention, eliminating the interfering pixels' negative influence and paying more attention to the same class pixels in one HSI patch.

For the proposed schemes, the idea has the ability to learn the correlations among input steps pixel by pixel instead of 3-D CNN, which convolves them together. Scheme-2 can get better results than Scheme-1, and the results are similar to SSAN. Local spatial sequences come up with local spatial information, allowing us to exploit the relationship of pixels in a neighboring area, which is ignored by CNN-based methods. Looking closely at Fig. 14(g) and (h), the proposed models lead to clearer boundaries. Moreover, it can obtain more accurate classification results in some details, such as the area marked by the white boxes.

2) Pavia University Dataset: Fig. 15 shows the classification results with different methods. The quantitative results for each method are also listed in Table VII.

From Table VII, we reach the same conclusion that the methods containing the local spatial features of the HSI patch have better classification accuracy than those classifiers just based on a single pixel's spectral information. By exploiting the local pixels' relationship, the 3-D CNN-based model is

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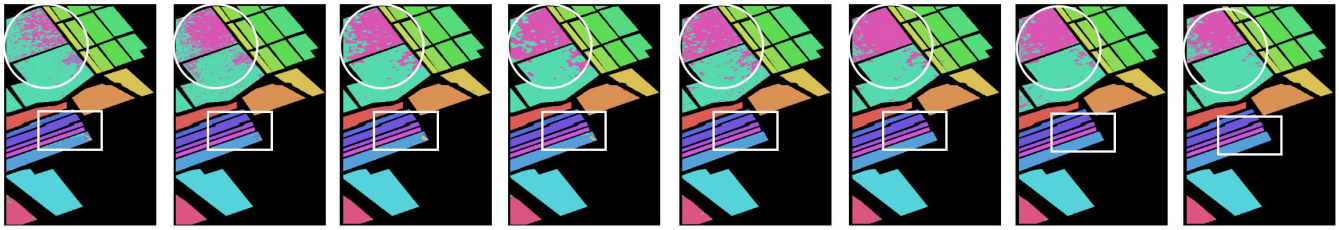


Fig. 16. Classification maps with different methods for the Salinas dataset. (a) Spectral CNN. (b) Deep RNN. (c) FCN. (d) 3-D CNN. (e) M3D-DCNN. (f) SSAN. (g) Proposed Scheme-1. (h) Proposed Scheme-2.

TABLE VIII
CLASSIFICATION ACCURACY (%) FOR THE SALINAS DATASET ON THE TEST SET

Classes	Methods							
	spectral CNN	Deep-RNN	3-D CNN	FCN	M3D-DCNN	SSAN	Scheme-1(ours)	Scheme-2(ours)
1	98.30	**99.88**	99.01	95.61	98.21	98.42	98.13	98.32
2	98.91	99.81	98.90	99.69	**99.96**	99.97	98.97	98.99
3	96.98	98.36	98.02	98.13	99.64	98.14	**100**	**100**
4	98.78	98.71	98.89	97.71	**99.78**	97.95	99.11	99.35
5	97.86	98.48	99.11	98.48	**99.46**	98.09	99.08	99.34
6	99.89	99.89	**100**	98.3	99.61	**100**	99.79	**100**
7	99.23	99.74	99.13	98.32	99.59	**99.94**	98.98	99.63
8	80.15	84.15	88.87	**94.91**	92.37	91.82	93.07	93.18
9	98.71	99.43	99.83	99.39	99.80	**99.97**	99.65	99.76
10	88.39	96.51	96.89	95.92	**98.41**	96.82	96.79	97.71
11	88.88	97.26	96.99	96.28	97.68	**99.24**	98.03	98.23
12	97.73	98.99	99.33	97.53	98.75	**99.47**	99.01	98.85
13	96.17	99.43	98.79	97.77	99.08	**100**	**100**	**100**
14	94.08	97.61	98.48	97.24	**98.73**	98.66	98.23	98.16
15	56.12	72.61	82.26	**90.42**	86.67	89.15	88.69	89.08
16	97.69	**99.01**	97.75	85.28	94.23	97.80	93.67	93.66
OA	89.46	92.95	95.06	94.71	95.83	97.23	97.19	**97.92**
AA	91.99	95.24	97.01	96.31	97.62	**97.84**	97.58	97.67
Kappa	0.87	0.92	0.94	0.94	0.94	**0.97**	**0.97**	**0.97**

more accurate than spectral CNN, deep RNN, and FCN. Take a look at classes "1," "2," and "7"; the 3-D CNN, M3D-DCNN, and SSAN models achieve obvious improvements. Especially, class "5" gets 100% accuracy.

Our proposed method achieves better performance among all the methods. Furthermore, the validity of Scheme-1 and Scheme-2 models is also proved. The highest classification accuracy is obtained by Scheme-2 model for classes "1," "3," "4," "6," "7," and "9." Moreover, OA, AA, and kappa are the highest among all the comparison methods. The kappa coefficient is even close to 1.

It is clear that the classification results of those spectral-based methods are very poor, as shown in Fig. 15(a) and (b). Using local area information, FCN, 3-D CNN, M3D-DCNN, and SSAN models can achieve cleaner results (marked by the white circles and boxes). By considering local features, wider receptive information is beneficial for training a more discriminative model. The proposed two models reduce the noise-like misclassifications, keep clear boundaries, and obtain better classification results in many small regions (marked by white circles). Overall, the proposed Scheme-2 model outperforms other methods.

3) Salinas Dataset: The classification results of the different methods have been shown in Fig. 16 and Table VIII, and the best results are marked in bold.

Due to this dataset's high homogeneity, the patch's size of all the methods is set as 7 for 3-D CNN, FCN, SSAN, and the proposed models. Other parameters are the same as those mentioned above. From Table VIII, we can find that the spectral-based methods, spectral CNN, and deep RNN, which directly input the spectral signatures into the model, cannot get satisfying results, especially the class "8" and "15." Most of them result from unbalanced training samples and the complex situation in spectral signatures. However, deep RNN directly considers the band-to-band correlation in one spectral signature and gets the best result in class "16." This result shows that RNN is good at dealing with the sequence benefiting from global dependence.

In the HSI, different class pixels may have similar spectral signatures; meanwhile, the same class pixels may have different big variations. Without considering the correlations from neighboring pixels, the classification results, as shown in Fig. 16(a) and (b), present noise-like misclassification. Moreover, all the other methods that consider spatial information have promising classification results. For example, take a look at Fig. 16(e) and (f), and the regions marked by a white circle are cleaner. Instead of noise-like misclassifications, object-like misclassifications can be seen. Moreover, look at the quantitative result of class "16"; by considering the spatial information, sometimes, the classification accuracy decreases to a certain extent. Our method achieves the best OA and kappa; especially, classes "3," "6," and "13" get 100% accuracy. Fig. 16 clearly shows that our methods can sustain more accurate and smoother classification results. It is obvious that

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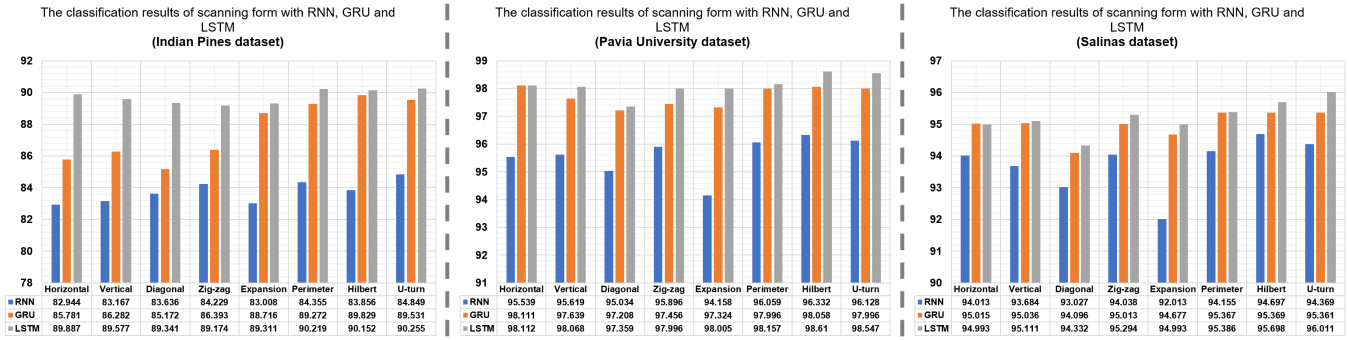


Fig. 17. Classification results of scanning forms with RNN, GRU, and LSTM on three datasets.

methods that use local spatial information result in many clean outlines of the regions. SSAN is one of the methods that take spatial attention into account also gets outstanding results. In many classes, it shows the best results. It means that spatial attention can decrease the negative influence of interfering pixels in one HSI patch. Compared with SSAN, our methods can get very similar results, and they are also only about 0.7% different in OA, as shown in Table VIII.

D. Other Analyses 1) to 8)

1) Performance on Different Scanning Forms: In our experiments, we extended and investigated different scanning forms for the classification results. As shown in Fig. 4, we have selected eight possible scanning forms: horizontal, vertical, diagonal, zig-zag, expansion, perimeter, Hilbert, and U-turn. Each scanning form can generate several scanning orders, such as Fig. 5. Hence, we first implement the experiments using one scanning order of each scanning form following Fig. 4. In the meanwhile, simple RNN, GRU, and LSTM units are deployed separately. Fig. 17 shows the results in three datasets.

From these figures, we can find that those scanning forms with nonadjacent connections between pixels got relatively lower classification accuracy, such as horizontal form and vertical form. In contrast, the scanning forms, perimeter form, Hilbert form, and U-turn form can get considerably higher accuracy than the other scanning forms. These three scanning forms highly follow our assumption that the scanning order should be totally spatially continuous. Compared with the perimeter scanning form, the Hilbert scanning and U-turn scanning forms, which arrange more nearest pixels alongside the central pixel in the sequence, can generally perform better. Meanwhile, the perimeter scanning form and the expansion scanning form are totally reversed, which means that they can be used to analyze the effects of the first step or last step that counts for the whole sequence by RNN. In the three datasets, the perimeter scanning form usually gets better results than the expansion scanning form, which explains that, in the RNN, the last step of the sequence generally plays a more important role to influence the final output of the RNN.

Comparing the diagonal form and the zig-zag form, we can find that the results from the zig-zag form seem better than the diagonal form. As shown in Fig. 4, the diagonal form contains more nonadjacent connections than the zig-zag form.

TABLE IX
PERFORMANCE ON COMBINING DIFFERENT SCANNING FORMS

	U-Turn	Horizontal	Vertical	OA(%)
Indian Pines	✓			**95.38**
		✓	✓	95.14
	✓	✓		94.97
	✓		✓	95.01
	✓	✓	✓	94.08
Pavia University	✓			**99.18**
		✓	✓	98.76
	✓	✓		98.91
	✓		✓	98.33
	✓	✓	✓	98.07
Salinas	✓			**97.92**
		✓	✓	97.33
	✓	✓		97.89
	✓		✓	97.67
	✓	✓	✓	97.11

This comparison proves that the appropriate scanning form should be highly spatially continuous.

2) Performance on Combining Different Scanning Forms: In this article, we have investigated eight scanning forms, and each form generally has eight scanning orders, except horizontal and vertical scanning forms. According to the previous experiments, U-turn scanning forms with eight scanning orders generally get the best results. Hence, we attempt to investigate the performance of combing different scanning forms briefly. The following parts are investigated: 1) U-turn scanning forms with eight scanning orders; 2) horizontal + vertical scanning forms (all eight scanning orders); and 3) horizontal or vertical (four scanning orders) + U-turn (eight scanning orders). The results are shown in Table IX.

We can find that, on the one hand, casually combining different scanning forms decreases the overall accuracy of the classifier. On the other hand, combining more scanning forms means the increment of training parameters, which shows that the combination is not a good tradeoff to better the classifier. That is why we just chose one scanning form (U-turn) to implement the subsequent experiments.

3) Effects of Different Feature Combination Manners: As we mentioned before, in one scanning form, each scanning order complements one image patch. Therefore, the appropriate combination of all features from possible scanning orders is necessary. We have investigated the current existing combination manners (SS, LLC, and FLC) in the community, as shown in Fig. 8. More experiments are implemented to find the

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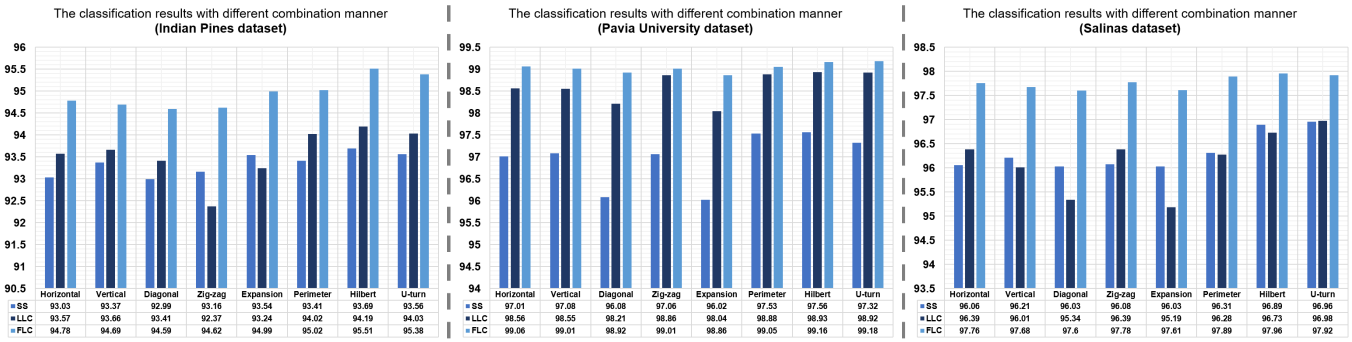


Fig. 18. Classification results of scanning forms with different combination manners on three datasets.

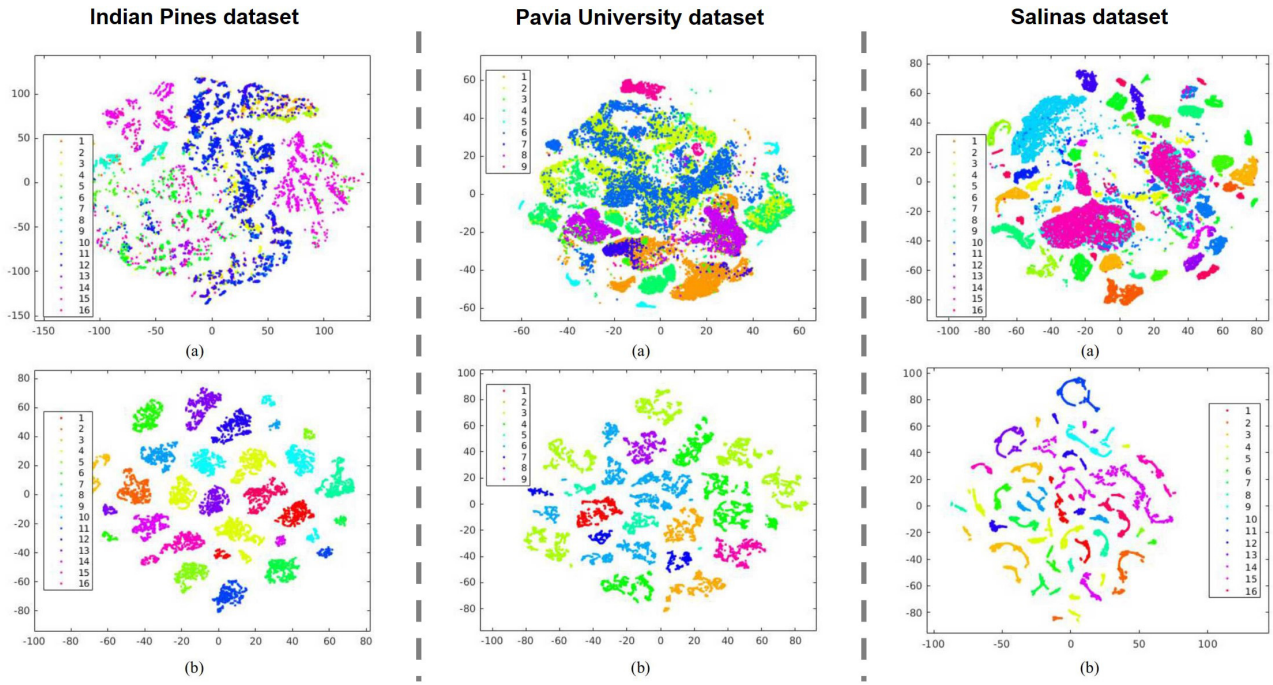


Fig. 19. t-SNE Visualizations on three datasets. (a) Original samples and (b) Learned feature representations by our method set by t-SNE.

optimal combination manner to deal with the complementary information among scanning orders. We have to note that eight complementary scanning orders are used in this experiment for each scanning form.

In this analysis, all of them are used for combination under LSTM. The classification results are shown separately in Fig. 18. The FLC combination manner always shows better results than the SS and LLC manners, proving that the correlative dependence among different scanning features is important for generating more discriminative representations. Compared with the LLC combination manner, FLC has shorter length (8) steps for the LSTM than the much longer length (hidden size $\times$ 8) in the LLC manner. It is a pretty long sequence for the LSTM to process, which directly influences the final results negatively. The SS combination manner directly sums up the weighted features together, which ignores the correlation among these features. Thus, generally, SS manner cannot get satisfying results.

From subsequent 4) to 8), the following analysis is based on Scheme-2 model with a U-turn scanning form.

4) Feature Visualizations by t-SNE: Deep-learning-based methods aim to create a new feature representation to replace the original input, known as representation learning. For example, most existing methods train one discriminate model from the input to a discriminative output representation in the HSI classification task. In contrast, spectral-based methods directly transform the original spectral signature of one pixel into another embedding space with new feature values. Then, new features are deployed against others for the final clustering. Spectral–spatial-based methods are trying to represent one pixel's information by integrating its neighbors' features.

To verify the feasibility of the proposed method in representative ability, we use the t-SNE [49] technique to visualize the distributions of original spectral signatures and learned representative features from our model. Hence, Fig. 19 shows the visualization results of the Indian Pines dataset, the Pavia University dataset, and the Salinas dataset, respectively.

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TABLE X

TOTAL LOSS OF KL-DIVERGENCE WITH THE USE OF DIFFERENT NUMBERS OF SCANNING SEQUENCES

	The use of number of scanning directions				
	0	1	2	4	8
Indian Pines	2.2780	1.3256	0.9935	0.8841	0.7846
Pavia University	3.7278	1.8250	1.4513	1.2117	1.0374
Salinas	2.3706	0.9871	0.9698	0.8211	0.7917

From these figures, two important issues should be clarified. First, in the results of t-SNE, some different color clusters are pretty close, but this kind of distance is meaningless for deciding whether they are similar or not. The internal mechanism of the t-SNE decides it. Second, we can find that some same color clusters are very far away. Because t-SNE is optimizing one nonconvex function, t-SNE focuses on learning to maintain the local structure and get a local minimum.

As shown in these figures, we can clearly see that our model can easily discriminate that learned feature representations of original pixels. Although the unbalanced samples are selected for visualization, the results have already been promising and satisfying. Furthermore, the new feature representations become clustering and discriminative by the t-SNE; several clusters are unified with the same color and become smoother.

Furthermore, we recorded the total loss of KL divergence [50] for each dataset using different numbers of scanning directions in Table X. We can find that the loss can intuitively decrease by increasing the number of scanning sequences. Thus, it verifies that the multiscanning strategy is feasible.

5) Effects of the Number of Scanned Pixel Sequences: In our experiments, we use the word "scanning" instead of "sorting" because the sorting operation may not consider the continuity between every pixel. For example, if we sort one 5×5 patch into one 1×25 sequence, we will get P_{25}^{25} sorting manners, and most of them are meaningless and redundant. Therefore, the scanning operation should help keep the structural continuity among one spatial patch, maintaining the locality and continuity between each pixel.

Data augmentation is generally deployed in the computer vision field for training a more discriminative network. The image can be augmented by rotation, mirroring, and flipping individually to increase the amount of input information for the image-based networks, such as CNNs. By changing the perspective, our proposed multiscanning strategy can also be seen as one of the data augmentation operations for generating sufficient sequences for sequence-based networks, such as RNNs. The augmented sequences will be used simultaneously for increasing the information amount.

More experiments are performed to investigate the effects of the number of used pixel sequences attributed to the system performance. For the Scheme-2 model, four pairs of spatial sequences are added into the training procedure pair by pair; two pixel sequences are in one pair. Fig. 20 shows the effects of numbers of input data for final classification results.

Increasing the number of input sequences improves the classification accuracy, proving that our methods and ideas are feasible. Meanwhile, we find that the accuracy of our approach

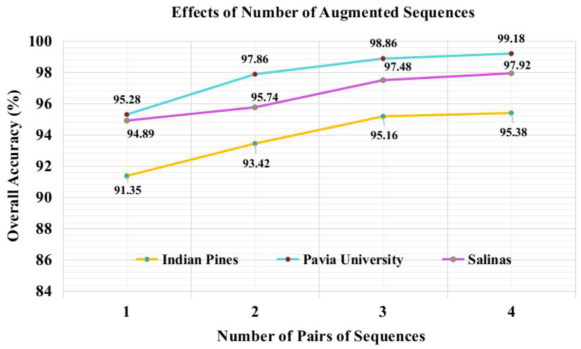


Fig. 20. Effects of the number of augmented data for three datasets.

TABLE XI

EFFECTS OF THE ADDING ATTENTION MECHANISM

	The use of attention layer			
	Scheme-1		Scheme-2	
	without	with	without	with
Indian Pines	94.33	95.13	94.57	95.38
Pavia University	98.51	98.99	99.01	99.18
Salinas	96.76	97.19	97.59	97.92

on the Salinas dataset does not increase much by increasing the number of sequences. Due to this dataset's high homogeneity and big size, one small HSI patch relatively always contains the same class pixels with a much higher possibility.

6) Effects of Using Attention Mechanism: Motivated by the attention mechanism, we have also added the attention layer into our models, as written in Section III. For different scanning orders, different weights should be allocated to count the importance of each scanning order, and these weights are generated by one single-layer neural network and one Softmax layer as formulated in (11) and (12). Hence, to verify the performance gain by the attention mechanism, we conducted more experiments. Moreover, the classification results (%) are shown in Table XI. These experiments are done with a patch size of 5 for the Indian Pines and Pavia University datasets and 7 for the SA dataset.

By adding the attention layer into the model, the network tends to allocate more importance for critical scanning order and reduce importance for others instead of allocating equally. Therefore, the outputs become more discriminative, resulting in higher classification accuracy. Both in Scheme-1 and Scheme-2, the accuracies increase by adding an attention layer. The improvements are significant, especially for the Indian Pines dataset; we assume that, due to the relatively small size of the Indian Pines dataset, the cropped patch with patch size five will contain more interfering pixels, which means that several scanned pixel sequences may have a negative influence on the training. The attention mechanism can allocate lower importance for these negative pixel sequences and contribute to better performance.

7) Comparison Between RNN and 1-D CNN for Dealing With Pixel Sequence: We propose processing pixel sequence using RNN (GRU and LSTM), which has the input channel of data spectral dimension C and sequence length of flattened pixels (patchsize $\times$ patchsize). Meanwhile, it can also

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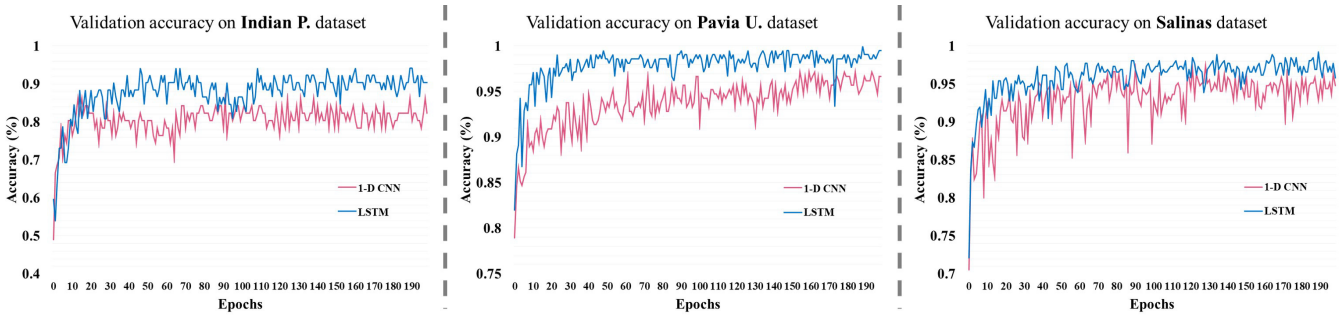


Fig. 21. Comparison of validation accuracy between RNN (LSTM) and 1-D CNN on three datasets for dealing with a pixel sequence.

TABLE XII
AVERAGE TRAINING, TESTING (SECOND) TIME, AND THE AMOUNT OF PARAMETERS OF EACH METHOD ON EACH DATASET

	Indian Pines			Pavia University			Salinas		
	Training(s)	Testing(s)	Params.	Training(s)	Testing(s)	Params.	Training(s)	Testing(s)	Param.
spectral CNN[12]	78.5	6.2	72,297	68.9	4.5	61,370	86.9	8.4	74,297
deep RNN[13]	874.1	12.9	1,828,561	736.2	10.4	1,664,458	898.3	16.4	1,833,425
FCN[18]	779.1	22.6	2,317,841	921.7	31.1	1,273,994	903.4	34.5	2,360,849
3-D CNN[21]	867.2	26.1	2,680,487	741.7	20.7	806,890	935.2	37.7	2,734,897
M3D-DCNN[35]	792.4	25.7	984,897	634.1	29.9	549,306	824.5	33.6	989,249
SSAN[39]	1834.2	67.6	103,847,643	1894.4	59.7	40,401,620	2923.6	68.9	109,614,811
Proposed	797.9	23.2	570,962	631.7	22.2	371,851	856.4	26.8	579,154

be processed by spectral–spatial 1-D CNN, which has the input channel of patchsize × patchsize and sequence length of spectral dimension C.

To verify the superiority of RNN, more experiments are implemented. We manually set the almost same number of parameters in these experiments with the proposed method for 1-D CNN. Moreover, we just set one feature extraction layer for both LSTM and 1-D CNN. The details of 1-D CNN settings are given in the following. The kernel size is set as 3. The number of kernels is 128. The kernel stride is 1. For the LSTM, we set one layer with 128 hidden sizes. Therefore, the experimental results are shown in Table XII. Furthermore, the comparisons of validation accuracy on three datasets are shown in Fig. 21.

The RNNs can better process the pixel sequence with fewer parameters than 1-D CNN, which demonstrates the superiority of RNN for dealing with the sequence-like data.

8) Training Time, Testing Time, and the Number of Parameters With Total Size: In this subsection, we show the training and testing time based on our RTX1080 GPU machine to verify the efficiency of all the methods. In this study, all the experiments are implemented with the Ubuntu18.04 operating system, Pytorch-1.1 toolbox. As shown in Table XIII, we can find that our method can take less training time and testing time with low computational efficiency compared with CNN-based methods, except spectral CNN, which just considers the spectral information. For those methods that consider the spatial information, the patch is set as 5, and the other parameters follow those references.

Furthermore, we can see that the amount of parameters in our method is largely less than those CNN-based spectral–spatial classifiers, which illustrates that our method can obtain better results with higher effectiveness. On the other hand, SSAN combines the 3-D CNN with self-attention to get good results, but it contains too many parameters, which seems not more efficient than ours.

TABLE XIII
COMPARISON BETWEEN RNN AND 1-D CNN FOR DEALING WITH PIXEL SEQUENCE

	1-D CNN		RNN (LSTM)	
	Params.	OA(%)	Params.	OA(%)
Indian Pines	491,281	81.267	416,618	90.324
Pavia University	164,874	96.214	116,607	97.676
Salinas	501,009	95.629	431,478	96.834

VI. CONCLUSION

We proposed a multiscanning strategy-based RNN for HSI classification. The main idea was to convert a local image patch using a multiscanning strategy into several complementary direction sequences with RNN to manage spatial–spectral information. Furthermore, we investigated the effects of different scanning forms on HSI classification results. Subsequently, we processed the four pairs of local direction sequences into Bi-RNN and then implemented data fusion by concatenation. Different concatenation manners (SS, LLC, and FLC) are also investigated for classification results. The multiscanning strategy can be considered one of the data augmentation operations that can be used to increase data volume for the RNN model. The proposed networks can extract the spectral information and spatial information in a patch of HSI, exploiting the spatial dependencies in the spatial patch with spectral information from each pixel.

Our experimental results on the three datasets proved that our method could obtain competitive results compared with the other well-known methods. In addition, as our approach uses fewer parameters, we believe that it is more efficient than 3-D CNN-based methods. Furthermore, we believe that our

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approach is a viable alternative to CNNs, and it can provide insights for future data analysis and research on only using RNN for constructing both spatial information and spectral information.

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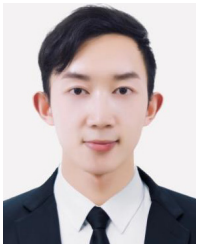
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